Conditional Generation with Classically Trainable IQP Circuits

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Conditional generation with a single compiled quantum circuit separates three questions that are often conflated: how the full output distribution responds to a condition, which training observables can be estimated classically, and whether the final circuit can be sampled classically. We study a fixed $H^{\otimes n} D_\theta H^{\otimes n}$ IQP circuit whose condition enters only through a frozen affine R_Y product input. Specializing established circuit-Fourier theory, every output probability lies in a finite multivariate signed-sum support. Refining the known IQP active-generator expansion for real product inputs, Pauli-Z responses retain only a filtered binary span and hence can have a smaller condition spectrum. The same product reference gives unbiased estimators for selected Pauli moments, an independent-copy Conditional MMD objective, and phase gradients, with explicit second-moment and resource bounds. Here "classically trainable" refers only to access to those quantities; it does not imply efficient output probabilities, classical sampling, optimizer convergence, or a favorable landscape. An adversarial primary-source audit found the probability theorem to be a direct specialization and the observable formula to be a short synthesis of prior results. The first binding novelty gate therefore failed. Under a frozen 400-run protocol, this decision blocked the planned 16-qubit PennyLane-Ising feasibility, tuning, primary, spectral, and ancilla branches; zero full training runs were reserved or executed. We report the complete derivations, executable checks, frozen benchmark, immutable kill tree, and stopped-work artifacts as a mechanism-and-boundary record. The result is an internally complete failure-path draft, not a submission-ready novelty claim.

Draft status (10 August 2026). The internal correctness audit passes and the theorem-novelty gate fails. No Ising model training was released. Independent quantum-information review and real author metadata are absent. This manuscript must not be submitted in its present form.

1 Introduction

Conditional generation asks one model to represent a family of probability laws $q_\theta(x \mid c)$ selected by an external condition c. In a quantum circuit Born machine, the most restrictive deployment version of this task is to keep one compiled gate sequence and one phase vector at every condition. Only the prepared input may change. This constraint is operationally sharper than sharing a parameterization whose effective gate angles are recomputed for each c, and much sharper than storing a separate circuit for every condition.

Input encodings already determine the accessible Fourier spectrum of a parameterized quantum model. Pauli rotation encodings give finite trigonometric response spaces, and affine or multi-dimensional encodings map their base frequencies to signed sums of encoding weights [1–3]. This observation constrains each output probability, but by itself supplies no efficient probability evaluator. Conversely, probabilistic simulation can estimate selected observables without strongly simulating the output law [4]. Recent IQP work makes this distinction concrete through active-generator Pauli estimators, automatic differentiation, and MMD objectives [5, 6].

The technical question in this paper is therefore a three-way boundary. We ask how a fixed circuit responds when c changes an affine product input, which smaller part of that response is visible to classically estimable Pauli observables, and which conclusions remain unavailable about final output sampling. The mechanism is shown in Fig. 1. Its input is $\bigotimes_i R_Y(b_i + w_i^{\mathsf{T}} c)|0\rangle$; all generators, their order, and θ remain byte-identical across conditions.

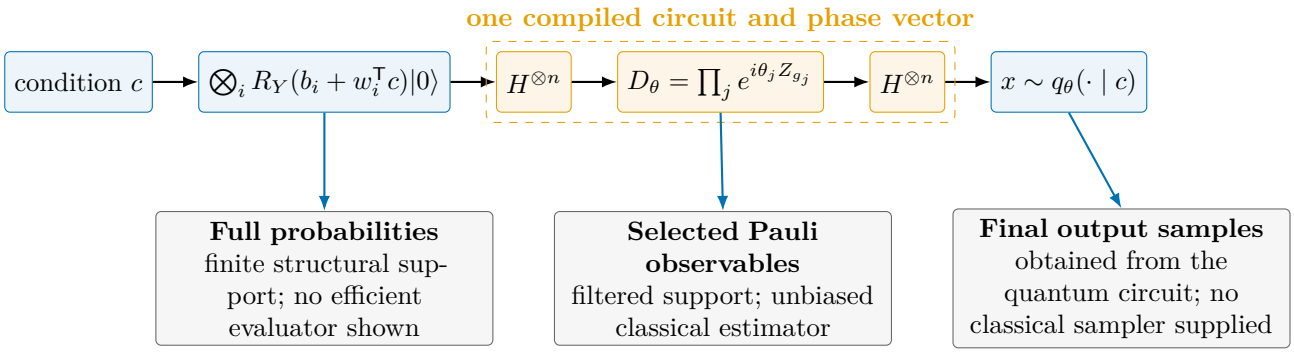


Figure 1: Fixed-circuit conditional generation and the three distinct objects studied in this paper. Only the product-state input changes with c; the compiled IQP gates and learned phases do not. A structural theorem for full probabilities is not an efficient probability or sampling algorithm.

Our analysis packages four finite-resource consequences. First, every output probability has support contained in a common multivariate signed-sum set. Second, conjugating a Pauli observable through the IQP block selects anticommuting generators; grouping their $\mathbb{F}_2$ span and using $\langle Y \rangle = 0$ on a real R_Y product input yields a sharper observable support. Third, the same representation gives bandwidth, derivative, and fixed-width non-density statements. Fourth, the product reference preserves unbiased selected-moment, Conditional MMD, and phase-gradient estimation. These are useful specializations and diagnostics, not claims of a new general Fourier or IQP-estimation framework.

The study was deliberately gated before empirical scale-up. A written novelty dossier compared every candidate theorem with primary sources, attempted counterexamples, and executable small-system checks. Although the internal algebra survived, no central result cleared the preregistered requirement of being more than a direct corollary or short synthesis. The binding gate consequently stopped a frozen 16-qubit PennyLane-Ising campaign before feasibility or model training. This adverse decision is the primary campaign result. It prevents preparatory exact targets, calibration, or older Gaussian experiments from being repurposed as evidence for an unexecuted Ising claim.

The contributions that remain supportable are:

1. an exact deployment contract for affine product-input conditioning of a fixed IQP circuit, together with a careful separation of probability response, observable estimation, and sampling;
2. a self-contained specialization of the known full response spectrum and an observable-specific active-span/real-input filter, with bandwidth, smoothness, and fixed-resource limitations;
3. explicit finite-variance estimators and operation counts for selected Pauli moments, sampled Conditional MMD, and phase gradients; and
4. an executable, immutable campaign protocol whose failed theorem gate provably blocks all 400 allocated full-training runs rather than silently recycling them.

2 Related work

Encoding spectra and expressivity. Gil Vidal and Theis characterize Pauli-encoding responses as multiaffine trigonometric functions and show how repeated affine encodings create signed frequency sums [1]. Schuld, Sweke, and Meyer derive accessible frequencies from encoding-Hamiltonian eigenvalue differences and treat multivariate inputs explicitly [2]. Casas and Cervera-Lierta study degrees of freedom and limitations in multidimensional Fourier series realized by quantum circuits [3]. Our full-probability statement takes a measurement projector as the observable and specializes those results to one affine R_Y input per qubit. It is not a priority claim.

IQP simulation boundaries. Commuting and IQP circuits have sharply different notions of classical simulation. Selected quantities can

admit probabilistic estimators even when strong output-probability evaluation is unavailable [4, 7]. Sampling-hardness results instead depend on specific circuit families and complexity conjectures [8, 9]. We use no hardness assumption and make no advantage claim. Efficient access to the training observables is kept logically separate from the final act of drawing an IQP output sample.

Observable-trained IQP models. IQPopt develops scalable classical optimization of IQP Pauli objectives and automatic differentiation [5]. Recio-Armengol, Ahmed, and Bowles give the active anticommuting-generator expansion, unbiased Pauli and MMD estimators, and a train-classically/deploy- quantum workflow [6]. Our narrow extension lets the efficiently sampled product reference depend on c and then evaluates their active expansion on a real affine product input. The resulting Y filter and condition-frequency map are convenient diagnostics; the novelty audit classifies them as a short synthesis.

3 Fixed-circuit conditional model

Let $g_j \in \mathbb{F}_2^n$ be binary generator masks, $Z_{g_j} = \bigotimes_i Z_i^{(g_j)_i}$, and define

$$D_\theta = \prod_{j=1}^{m} e^{i\theta_j Z_{g_j}}, \qquad U_\theta = H^{\otimes n} D_\theta H^{\otimes n}. \tag{1}$$

All g_j, their order, and θ are independent of the condition. For $c \in \mathbb{R}^d$, the fixed encoding is

$$\alpha_i(c) = b_i + w_i^{\mathsf{T}} c, \qquad |\psi(c)\rangle = \bigotimes_{i=1}^{n} R_Y(\alpha_i(c))|0\rangle, \tag{2}$$

where $W \in \mathbb{R}^{n\times d}$ has rows w_i^{T}. Neither W nor b is learned. The theorem treats them as arbitrary fixed coefficients; the campaign chooses one map from target-family-only interval calibration and shares it across all coupling instances. The conditional output law is

$$q_\theta(x \mid c) = |\langle x|U_\theta|\psi(c)\rangle|^2. \tag{3}$$

The deployment distinction is structural, not terminological. A fixed-input model changes Eq. (2) and keeps every effective IQP phase fixed. A condition-modulated-phase baseline keeps 136 shared coefficients but maps c into effective gate phases, so its compiled numerical phases vary. A no-condition model receives one input for all labels. Separate models use one independent 136-parameter vector per training condition and are capacity references, not conditional generators. The implementation compares serialized deployment signatures to enforce these distinctions.

4 Finite-resource response structure

The following statements are included for completeness and diagnostic value. Proposition 1 is a direct specialization of established circuit-Fourier results [1, 2]; it is not presented as new.

Proposition 1 (Full-probability support)**.** *For any fixed post-input unitary and any output x, there are coefficients $Q_{x,s}$ such that*

$$q(x \mid c) = \sum_{s\in\{-1,0,1\}^n} Q_{x,s} \exp\left(i \left[\sum_i s_i w_i\right]^{\mathsf{T}} c \right). \tag{4}$$

Consequently

$$\begin{aligned} &\operatorname{supp} \widehat{q}_x \subseteq \Omega_{\text{full}}(W), \\ &\Omega_{\text{full}}(W) := \{\textstyle\sum_i s_i w_i : s_i \in \{-1,0,1\}\}, \end{aligned} \tag{5}$$

with at most 3^r distinct frequencies when r rows of W are nonzero.

The inclusion can be strict: repeated rows collide, special phases cancel coefficients, and zero rows add no frequency. Moreover, Eq. (4) does not make a coefficient efficiently computable. The statement is about the response class, not strong simulation.

For a measured mask $a \in \mathbb{F}_2^n$, write $X_a = \bigotimes_i X_i^{a_i}$ and define the active set and span

$$\begin{aligned} \mathcal{A}(a) &= \{j : a \cdot g_j = 1 \pmod 2\}, \\ \mathcal{H}_a &= \operatorname{span}_{\mathbb{F}_2}\{g_j : j \in \mathcal{A}(a)\}. \end{aligned} \tag{6}$$

Proposition 2 (Observable active-span filter)**.** *Let*

$$f_a(c) = \langle\psi(c)|U_\theta^\dagger Z_a U_\theta|\psi(c)\rangle. \tag{7}$$

Then

$$f_a(c) = \sum_{\substack{h\in\mathcal{H}_a \\ h\wedge a=0}} \kappa_h(\theta) \prod_{i:a_i=1} \cos\alpha_i(c) \prod_{i:h_i=1} \sin\alpha_i(c), \tag{8}$$

where κ_h is the sum of the active-generator subset coefficients whose binary XOR equals h. Write

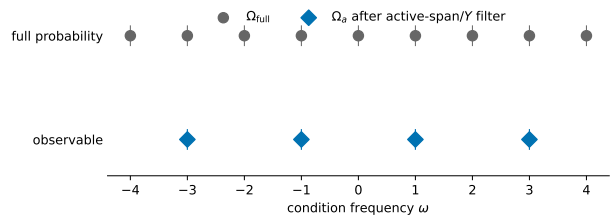


Figure 2: Illustrative scalar supports. The full probability support is a common structural upper bound; an observable can retain only a filtered subset. Frequencies and coefficients depend on the encoding and circuit, and permitted components can still vanish.

$S_{a,h} = \operatorname{supp}(a) \cup \operatorname{supp}(h)$. *Then*

$$\operatorname{supp} \widehat{f}_a \subseteq \bigcup_{\substack{h \in \mathcal{H}_a \\ h \wedge a = 0}} \Omega_{a,h},$$

$$\Omega_{a,h} := \left\{ \textstyle\sum_{i \in S_{a,h}} \sigma_i w_i : \sigma_i \in \{-1, +1\} \right\}. \quad (9)$$

Only generators anticommuting with X_a survive conjugation through D_θ. Expanding them produces $X_a Z_h$; after the first Hadamards this is $Z_a X_h$. Any overlap $h \wedge a$ introduces a local Y, whose expectation vanishes on the real R_Y product state. This derivation refines the known active-subset expansion [6]; the adverse audit does not assign it independent theorem-level novelty.

Let $B_k = \sum_i |W_{ik}|$. Trace-norm differentiation of the product input gives the following finite-resource consequences.

Corollary 3 (Bandwidth and fixed-width limitation)**.** *For any fixed effect M with $\|M\|_\infty \le 1$ and $F(c) = \operatorname{Tr}[MU\rho(c)U^\dagger]$,*

$$|\partial^\gamma F(c)| \le \prod_{k=1}^{d} B_k^{\gamma_k}. \quad (10)$$

Along direction v, the rth derivative is at most $(\sum_i |w_i^\mathsf{T} v|)^r$. For f_a, both sums restrict to the active observable lightcone. For fixed n, W, b, the resulting probability-valued maps are not dense in the continuous probability families on any compact set with nonempty interior.

The non-density statement follows because the finite-dimensional span of Eq. (5) is closed: a two-outcome target containing a cosine at any frequency outside that span has a positive uniform approximation floor. It says nothing about a sequence whose width, re-upload depth, or encoding weights grow. Likewise Eq. (10) describes response regularity, not gradient magnitude or optimization landscapes.

5 Classical estimation of the training objective

Set $|\chi(c)\rangle = H^{\otimes n}|\psi(c)\rangle = \sum_z \phi_c(z)|z\rangle$ and $\pi_c(z) = |\phi_c(z)|^2$. The R_Y input makes both objects products of local terms, so a reference string can be sampled and its amplitude evaluated in $O(n)$ operations. Define

$$\Phi_\theta(z) = \sum_j \theta_j (-1)^{g_j \cdot z} \quad (11)$$

and, with value zero outside the support of π_c,

$$R_{a,c,\theta}(z) = \frac{\phi_c(z \oplus a)}{\phi_c(z)} e^{i[\Phi_\theta(z \oplus a) - \Phi_\theta(z)]}. \quad (12)$$

Proposition 4 (Product-reference estimator)**.** *For $z \sim \pi_c$,*

$$\mathbb{E} \operatorname{Re} R_{a,c,\theta}(z) = f_a(c), \qquad \mathbb{E}|R_{a,c,\theta}(z)|^2 \le 1. \quad (13)$$

Only $j \in \mathcal{A}(a)$ enter the phase difference. Since π_c is independent of θ, termwise differentiation is unbiased; each active gradient component has second moment at most four and every inactive component is exactly zero for that observable.

The estimator is finite-variance but not pointwise bounded: a rare reference outcome can carry a large amplitude ratio. The ordinary S-sample mean has RMSE at most $S^{-1/2}$ and admits a Chebyshev guarantee. A median-of-means wrapper gives logarithmic failure-probability dependence but is not generally unbiased. A Hoeffding bound for the raw ratio would be unjustified.

For the Hamming kernel $k_\sigma(x, y) = \exp[-d_H(x, y)/(2\sigma^2)]$, a product distribution over binary masks represents the kernel as a mixture of Pauli characters. With $p_\sigma = (1 - e^{-1/(2\sigma^2)})/2$ and $q_\sigma(a) = p_\sigma^{|a|}(1 - p_\sigma)^{n-|a|}$,

$$k_\sigma(x, y) = \sum_{a \in \{0,1\}^n} q_\sigma(a)(-1)^{a \cdot x}(-1)^{a \cdot y}. \quad (14)$$

This is the product mask law used by the IQP MMD estimator [6]. A naive square of one estimated character mean is biased; independent cross-products or the corresponding U-statistic yield an unbiased squared discrepancy [10]. Averaging the three frozen bandwidths $\sigma \in \{1, 2, 4\}$

and balancing conditions produces the sampled Conditional MMD objective. Because masks, target minibatches, and product- reference samples are phase independent, autodifferentiating that finite sample expression gives an unbiased phase-gradient estimator.

Operational definition. In this paper, *classically trainable* means that the declared polynomial mixture of Pauli moments, its sampled Conditional MMD loss, and phase gradients are accessible to a classical estimator with polynomially declared budgets. It does not mean efficient arbitrary output probabilities, efficient classical output sampling, guaranteed convergence, nonvanishing gradients, absence of barren plateaus, or quantum advantage.

With local amplitudes cached, one reference sample costs $O(n)$ to draw, $O(|a|)$ for its flip ratio, and $O(\sum_{j\in\mathcal{A}(a)}|g_j|)$ for the active phase. For C conditions, R sampled observables, and S shared reference samples per condition, the direct batched cost is

$$O\left(CS\left[n+\sum_{r=1}^{R}\left(|a_r|+\sum_{j\in\mathcal{A}(a_r)}|g_j|\right)\right]+Cnd\right). \tag{15}$$

Writing a dense m-component gradient necessarily costs $\Omega(m)$. These counts separate qubit number, generator growth, condition count, and condition dimension; they are not claims about landscape trainability.

5.1 Synthetic scaling along declared resource axes

A zero-training-run audit used frozen synthetic parameters to vary qubits, generators, condition count, and condition dimension on both a bounded-degree $O(n)$ generator track and a Complete 2-local $O(n^2)$ track. Its 21 cases cover $n \in \{4, 8\}$, 8–36 generators, $K \in \{1, 4, 16\}$ conditions, and $d \in \{1, 2, 4\}$ condition dimensions, using 256 reference samples in each of three independent repeats. Fourteen $n = 4$ cases admit exact statevector references; their worst expectation RMSE was 0.0583 and worst active-coordinate gradient RMSE was 0.128. The seven $n = 8$ cases use repeated-estimator consistency instead of an exact reference in this audit; their worst consecutive-repeat RMSE was 0.0738 for the expectation and 0.130 for active gradients. Per-case wall time, process peak RSS, the Python/JAX/device environment, exact invocation, code revision, and source digests are archived. Timing and memory are host dependent; deterministic inputs, draws, numerical summaries, and their digest are separated from them. These are estimator error and resource measurements, not evidence about raw gradient magnitude, optimizer convergence, or landscape trainability.

6 Frozen Ising campaign and kill tree

The empirical protocol was frozen before model results so that an adverse gate could not be repaired by changing targets, thresholds, or run allocation. It uses the official periodic random-positive 4×4 PennyLane-Ising construction [11, 12]. For each coupling instance we exactly enumerate all $2^{16} = 65{,}536$ strings, with energy $E(s) = -\frac{1}{2}s^{\mathsf{T}}Js$ and target $p_\beta(s) \propto e^{-\beta E(s)}$. Seed 666 anchors the released generator; seven additional coupling seeds and aligned data seeds are frozen.

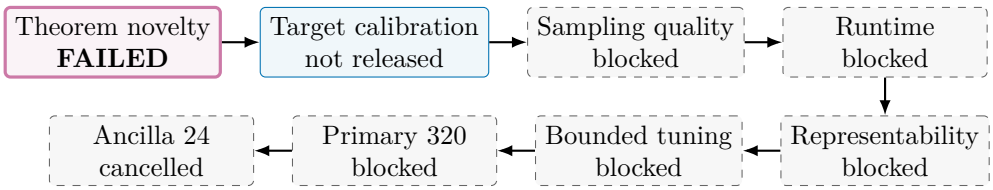
Target-only calibration screened declared intervals using exact adjacent-beta TVD, normalized entropy, and endpoint concentration. The widest eligible interval was $[0.15, 0.55]$, with five training points including $\beta = 1/3$ and four adjacent held-out midpoints. This artifact and eight exact target archives were generated while the theorem audit was open. They are preparatory evidence only: the target-calibration gate was never released because its predecessor failed.

The run ledger is content addressed. A run identity includes the campaign version, stage, coupling and data seeds, condition scope, model family, phase-feature candidate when applicable, ancilla rung, optimizer seed, and manifest digest. A terminal result must additionally record code revision, environment, exact invocation, timestamps, runtime, peak memory, parameter accounting, dataset hashes, estimator randomness, checkpoint provenance, raw metrics, and completion state. Reservations and results are immutable; duplicate or undeclared identities are rejected.

The full allocation is 20 feasibility/representability, 36 tuning, 320 primary, and 24 conditional ancilla runs. These are independent ceilings, not a fungible total. Monotonic

Table 1: Frozen, unexecuted Ising mechanism benchmark. Held-out conditions are inaccessible to calibration, tuning, feature selection, and checkpoint selection. Full-training-run use is zero because the first gate failed.

Component	Frozen rule	Status
Targets	8 periodic 4×4 instances; exact 2^{16} truth	Preparatory only
Input	One target-family-calibrated affine R_Y map, shared by all instances	Declared
Ansatz	Complete 2-local IQP; 16 qubits, 136 phases	Declared
Controls	Phase-modulated, no-condition, and separate models	Declared
Data	5,000 Training; 2,000 disjoint Validation strings	Not certified
Objective	Balanced batches; $2{,}000 \times 2{,}000$ estimator budget	Not run
Primary	$8 \times 5 \times (3+5) = 320$ full training runs	Blocked
Statistics	Exact held-out TVD; paired hierarchical bootstrap	Blocked
Ancillas	$m \in \{3, 5\}$ and matched-capacity controls	Cancelled



Unused allocations are non-transferable. A failed predecessor is terminal; later stages cannot be released by changing thresholds or reassigning runs.

Figure 3: Binding campaign kill tree and realized state. Preparatory exact targets do not bypass a failed predecessor. No feasibility, representability, tuning, primary, spectrum, or ancilla model run was launched.

gate records ensure that a theorem failure makes every downstream reservation ineligible. Tiny exact fixtures exercise the same transitions, model relations, leakage checks, result schemas, and statistics without consuming campaign runs.

7 Evidence and adverse novelty decision

The internal dossier contains exact assumptions, complete proofs, a primary-source matrix, counterexample attempts, executable checks, and a claim-evidence registry. Twenty-one focused theorem tests cover signed-sum enumeration, frequency collisions, exact small-circuit Fourier support, active-generator parity, binary spans, the real-input Y filter, joint-XOR survivors, inactive-phase cancellation, and bandwidth calculations. These checks support correctness only; they cannot establish novelty.

The combined active-span formula was not found verbatim, but absence of an identical displayed equation is not evidence of a central conceptual advance. Its ingredients and cancellation mechanism are explicit in the closest IQP work, and the remaining step is a short Pauli-algebra/Fourier synthesis. The novelty criterion therefore fails even though the statements survive the present internal checks.

Two complete sampler-development attempts preceded the final novelty verdict. With 2,000 burn-in sweeps and thinning intervals of 10 or 16, each missed the frozen per-chain energy-ESS threshold in 2 of 40 Validation cells; all other exact-reference moment, correlation, and low-weight-character checks passed. A more conservative 32-thinning attempt was interrupted immediately when the theorem verdict arrived, after six cells. These archived attempts neither pass the sampling gate nor consume full training runs. No model saw their samples.

One older 19-qubit Gaussian Hamming-profile study explains why the structured untwirled Ising benchmark was proposed. Across ten optimizer seeds in its archived protocol, the shared angle-conditioned model had mean-condition profile TVD 0.096 and Bitstring TVD 0.314, compared with 0.365 and 0.482 for no condition. The gap between profile and Bitstring fidelity motivated

Table 2: Closest-primary-result audit. "Delta" is the strongest accurate difference found, not a novelty claim. The binding requirement was at least one correct central theorem that was not a direct corollary or short synthesis.

Candidate	Closest primary result	Residual delta	Verdict
Probability support	Gil Vidal–Theis; Schuld et al.	Projector packaging	Specialization
Bandwidth	Gil Vidal–Theis; Casas et al.	Explicit fixed-W counterfamily	Corollary
Observable span/filter	Recio-Armengol et al.	Real-input evaluation	Short synthesis
Reference estimator	Van den Nest; IQPopt	Resource-axis accounting	Specialization

Table 3: Full-training-run ledger at the stop decision. Allocated runs remain unspent and cannot be moved to another pool.

Pool	Cap.	Reserved	Done
Feasibility/representation	20	0	0
Tuning	36	0	0
Primary	320	0	0
Ancilla	24	0	0
Total	400	0	0

exact untwirled evaluation on a non-shell target. It is historical motivation, not evidence for any of the blocked Ising claims; the full legacy audit is confined to the supplement.

8 Discussion

The fixed-input deployment contract remains conceptually useful. It rules out condition-dependent gate reprogramming and makes the source of conditional capacity explicit. The response calculation then separates a common full probability support from an observable-specific active span, while the product reference isolates exactly which training quantities remain classically estimable. This vocabulary prevents a finite-spectrum expressivity statement from being mistaken for strong simulation and prevents autodifferentiation from being mistaken for a landscape guarantee.

The same clarity exposes why the intended theory-led paper cannot proceed. The full support is a direct instance of general circuit-Fourier theory. The IQP refinement is useful bookkeeping but combines ingredients already explicit in the active-generator literature. Smoothness, fixed-width non-density, and the estimator follow by short consequences or specialization. Running 320 primary fits would at best supply evidence for a mechanism whose proposed theoretical headline had already failed the campaign's significance rule. Stopping is therefore not missing experimental work; it is the prescribed outcome of the experimental design.

No inference can be made about Ising representability, benefit retention, held-out interpolation, unresolved target spectral mass, or ancilla marginalization. Likewise, the analysis does not show conditional universality, BQP universality, efficient arbitrary probabilities, efficient classical sampling, optimizer convergence, immunity to barren plateaus, classical superiority, hard sampling for the finite instances, or quantum advantage. The exact 16-qubit targets and calibrated grid are protocol- development artifacts, not a substitute for the blocked comparison matrix.

A new campaign version could be justified only by a materially different central result. Plausible directions include a topology-dependent approximation lower bound using the active-generator code structure, algebraic constraints on attainable observable coefficients beyond frequency support, or a target-spectral excess-risk bound that predicts an independently observed interpolation boundary. Any candidate must undergo a new primary-source and proof audit, followed by independent specialist review, before model training is released.

9 Conclusion

We specified conditional generation with one fixed IQP circuit and an affine product-state input, then separated the resulting probability response from the selected observables used for training and from final output sampling. Established Fourier theory gives the finite full-probability support; the known IQP active expansion plus the real-input filter gives a sharper Pauli response; and a product-reference ratio gives finite-variance access to the declared Pauli, Conditional MMD, and phase-gradient quantities.

The strongest empirical fact in the frozen

campaign is that no full training run occurred. An internally correct theorem package failed the binding novelty test, so the immutable protocol blocked every downstream Ising and ancilla stage. That stop preserves the distinction between an explanatory synthesis and a publishable central advance. The present package is a reproducible mechanism-and-boundary record; it is not ready for submission to *Quantum* without a genuinely new result, a new campaign version, real author metadata, and independent expert review.

A Complete analytical details

This supplement gives the complete internal derivations behind Sec. 4 and Sec. 5. They are recorded for correctness and reproducibility. As established by the closest-primary-result audit in Table 2, the full-probability result is a direct specialization of circuit-Fourier theory and the observable result is a short synthesis of the known IQP active-generator expansion with the real-input filter. The proofs below are not presented as priority claims.

A.1 Proof of Proposition 1

The density operator of one encoded qubit is

$$\rho_i(\alpha_i) = R_Y(\alpha_i)|0\rangle\langle 0|R_Y(\alpha_i)^\dagger = \frac{1}{2}\left(I + \sin\alpha_i X + \cos\alpha_i Z\right). \tag{16}$$

Every matrix entry of Eq. (16) is a linear combination of the three angle frequencies $-1, 0, +1$. Tensoring the local states therefore gives fixed matrices A_s such that

$$\rho(c) = \bigotimes_i \rho_i(\alpha_i(c)) = \sum_{s\in\{-1,0,1\}^n} A_s e^{i\sum_i s_i\alpha_i(c)}. \tag{17}$$

Substituting $\alpha_i(c) = b_i + w_i^\mathsf{T} c$ absorbs the bias into the coefficient:

$$\rho(c) = \sum_s \left(A_s e^{i\sum_i s_i b_i}\right) e^{i(\sum_i s_i w_i)^\mathsf{T} c}. \tag{18}$$

For a fixed unitary U and outcome x,

$$q(x \mid c) = \mathrm{Tr}\left[U^\dagger|x\rangle\langle x|U\rho(c)\right] \tag{19}$$

is a fixed linear functional of $\rho(c)$, so it introduces no new frequency. There are at most three choices per nonzero row of W. Zero rows, coincident signed sums, and coefficient cancellations only shrink the realized set, which proves Eqs. (4)–(5).

The proof does not use the IQP structure. It also does not give an efficient method for evaluating the trace coefficient $Q_{x,s}$; exact evaluation can still scale as 2^n.

A.2 Proof of Proposition 2

The final Hadamards map the measured operator to X_a. A diagonal generator with $a \cdot g_j = 0 \pmod 2$ commutes with X_a and cancels between $D_\theta^\dagger$ and D_θ. For an active generator,

$$e^{-i\theta_j Z_{g_j}} X_a e^{i\theta_j Z_{g_j}} = X_a e^{2i\theta_j Z_{g_j}}. \tag{20}$$

Since all diagonal generators commute,

$$D_\theta^\dagger X_a D_\theta = X_a \prod_{j\in\mathcal{A}(a)} \left[\cos(2\theta_j) I + i\sin(2\theta_j) Z_{g_j}\right]. \tag{21}$$

Expanding Eq. (21) over active subsets S produces the binary mask $h = \bigoplus_{j\in S} g_j$. Grouping subsets with equal h gives

$$\kappa_h(\theta) = \sum_{\substack{S\subseteq\mathcal{A}(a)\\ \bigoplus_{j\in S} g_j = h}} i^{|S|} \prod_{j\in S} \sin(2\theta_j) \prod_{j\in\mathcal{A}(a)\setminus S} \cos(2\theta_j). \tag{22}$$

Conjugating $X_a Z_h$ through the first Hadamards yields $Z_a X_h$. The input is a product with local moments

$$\langle X_i\rangle = \sin\alpha_i, \qquad \langle Z_i\rangle = \cos\alpha_i, \qquad \langle Y_i\rangle = 0. \tag{23}$$

When $h \wedge a \neq 0$, at least one local factor is proportional to Y and the complete product expectation vanishes. Otherwise it factors into the sine/cosine product of Eq. (8).

For completeness, every subset contributing to a surviving h has even cardinality:

$$|S| \pmod 2 = \sum_{j\in S} a\cdot g_j = a\cdot \bigoplus_{j\in S} g_j = a\cdot h = 0. \tag{24}$$

Thus the surviving $i^{|S|}$ and κ_h are real. Expanding each local sine or cosine into its two exponentials proves Eq. (9).

An immediate union lightcone is

$$L(a) = \mathrm{supp}(a) \cup \bigcup_{j\in\mathcal{A}(a)} \mathrm{supp}(g_j). \tag{25}$$

The response is independent of all encoding rows outside $L(a)$, but the binary-span/Y filter can be strictly tighter than this union.

A.3 Derivative bounds

Every derivative of Eq. (16) with respect to its angle has trace norm one. For positive derivative order it has the form

$$\frac{1}{2}[u_r(\alpha_i)X + v_r(\alpha_i)Z], \qquad u_r(\alpha_i)^2 + v_r(\alpha_i)^2 = 1, \tag{26}$$

whose eigenvalues are $\pm 1/2$; at order zero ρ_i itself has trace norm one. Apply the requested condition-coordinate derivatives successively to $\rho(c) = \bigotimes_i \rho_i(\alpha_i(c))$. The product rule gives an ordered sum of tensor terms, each of trace norm one. For coordinate k, the absolute coefficients sum to $B_k = \sum_i |W_{ik}|$, hence

$$\|\partial^\gamma \rho(c)\|_1 \le \prod_k B_k^{\gamma_k}. \tag{27}$$

Unitary invariance and Hölder's trace inequality yield

$$|\partial^\gamma F(c)| \le \|M\|_\infty \|\partial^\gamma \rho(c)\|_1, \tag{28}$$

which proves Eq. (10). Replacing each coordinate coefficient by $w_i^\mathsf{T} v$ proves the directional form. For an observable response, Proposition 2 removes every qubit outside $L(a)$ before the same argument is applied.

A.4 Fixed-width non-density

Fix n, W, b and a compact condition domain $C \subset \mathbb{R}^d$ with nonempty interior. Let

$$V_\Omega = \mathrm{span}\{c \mapsto e^{i\omega^\mathsf{T} c} : \omega \in \Omega_{\mathrm{full}}(W)\} \subset C(C;\mathbb{C}). \tag{29}$$

This finite-dimensional subspace is closed in the uniform norm. Choose $\omega_\star$ with neither sign in the finite support. Distinct real-vector exponentials are linearly independent on an open set, so $g(c) = \cos(\omega_\star^\mathsf{T} c) \notin V_\Omega$ and

$$\delta = \inf_{v\in V_\Omega} \|g - v\|_\infty > 0. \tag{30}$$

For $0 < \epsilon \le 1/2$, define a valid two-outcome family

$$p(x_0 \mid c) = \tfrac{1}{2} + \epsilon g(c), \qquad p(x_1 \mid c) = \tfrac{1}{2} - \epsilon g(c). \tag{31}$$

The constant function lies in V_Ω, whereas Proposition 1 puts every $q(x_0 \mid \cdot)$ in V_Ω. Consequently

$$\sup_{c \in C} |q(x_0 \mid c) - p(x_0 \mid c)| \ge \epsilon\delta > 0. \tag{32}$$

This proves the fixed-resource limitation. It does not apply to a sequence in which the support itself grows.

A.5 Proof of Proposition 4

Expanding the Pauli expectation in the diagonal basis gives

$$f_a(c) = \sum_z \phi_c(z)^* \phi_c(z \oplus a) e^{i[\Phi_\theta(z \oplus a) - \Phi_\theta(z)]}. \tag{33}$$

On the support of π_c, multiplication of Eq. (12) by $\pi_c(z) = \phi_c(z)^*\phi_c(z)$ reproduces the summand in Eq. (33). Terms with zero denominator already vanish in the original sum, so the zero convention is harmless. The quantum expectation is real, proving unbiasedness of the real part.

The second moment is

$$\mathbb{E}_{z \sim \pi_c} |R(z)|^2 = \sum_{z : \phi_c(z) \neq 0} |\phi_c(z \oplus a)|^2 \le 1. \tag{34}$$

For $\chi_j(z) = (-1)^{g_j \cdot z}$, $\chi_j(z \oplus a) = \chi_j(z)(-1)^{g_j \cdot a}$. Therefore

$$\Phi_\theta(z \oplus a) - \Phi_\theta(z) = -2 \sum_{j \in \mathcal{A}(a)} \theta_j \chi_j(z). \tag{35}$$

The sample law is phase independent and the sum is finite, so differentiation commutes with expectation. Explicitly,

$$\partial_{\theta_j} R = \begin{cases} -2i\chi_j(z) R, & j \in \mathcal{A}(a), \\ 0, & j \notin \mathcal{A}(a). \end{cases} \tag{36}$$

Multiplying the second moment by four proves the active gradient bound.

A.6 Unbiased finite-sample Conditional MMD

For one bit $z \in \{0, 1\}$ and $p_\sigma = (1 - e^{-1/(2\sigma^2)})/2$,

$$\mathbb{E}_{A \sim \mathrm{Bern}(p_\sigma)} (-1)^{Az} = \begin{cases} 1, & z = 0, \\ e^{-1/(2\sigma^2)}, & z = 1. \end{cases} \tag{37}$$

Independence across bits, with $z = x \oplus y$, gives Eq. (14) and the product mask law $q_\sigma(a) = p_\sigma^{|a|}(1 - p_\sigma)^{n-|a|}$. Thus each mask selects one Pauli character and averaging masks recovers the Hamming kernel.

Fix one condition, bandwidth, and Fourier mask. For independent target characters $t_r = (-1)^{a \cdot x_r}$ and independent real estimator values $f_s = \mathrm{Re}\, R_{a,c,\theta}(z_s)$, define

$$\widehat{\Delta}^2_{a,c} = \frac{1}{N(N-1)} \sum_{r \neq r'} t_r t_{r'} + \frac{1}{S(S-1)} \sum_{s \neq s'} f_s f_{s'} - \frac{2}{NS} \sum_{r,s} t_r f_s. \tag{38}$$

Independence makes the three terms estimate the target square, model square, and cross-product without self-products, so

$$\mathbb{E}\widehat{\Delta}^2_{a,c} = \Big(\langle Z_a\rangle_{p_c} - \langle Z_a\rangle_{q_\theta(\cdot|c)}\Big)^2 . \tag{39}$$

Averaging Eq. (38) over the declared condition-balanced minibatch, three bandwidths, and Fourier masks is an unbiased Conditional MMD estimator. The stochastic value can be negative. Clipping it or taking its square root would alter the estimand. Since every source of randomness is phase independent, differentiating this same independent-copy expression is unbiased.

B Counterexample and theorem-check dossier

The executable theorem checks deliberately target overstatements that a surface-level derivation can miss.

1. **Structural support need not be attained.** For one qubit and a special IQP phase, an output expectation can be constant even though Ω_{full} permits nonzero frequencies.

2. **Repeated rows collide.** Three scalar rows equal to one produce only the seven frequencies $-3, -2, \ldots, 3$, not 27 distinct components.

3. **The union lightcone can be loose.** With $a = 100$ and $g = 110$, the active union contains qubit two, but the nontrivial term contains a local Y and the exact response is independent of its angle.

4. **Active generators can survive jointly.** For $a = 100$, $g_1 = 110$, and $g_2 = 101$, each one-generator term is removed, while their XOR 011 is disjoint from a and yields a joint $\cos\alpha_1 \sin\alpha_2 \sin\alpha_3$ term.

5. **Inactive phases cancel exactly.** Varying a generator that commutes with X_a leaves the exact response and its phase derivative unchanged.

6. **The amplitude ratio is not pointwise bounded.** Near a zero local reference amplitude, a rare sampled ratio can diverge even though its second moment remains at most one.

7. **Fixed width misses continuous families.** A cosine at a frequency outside the finite support gives the explicit uniform approximation floor used above.

The regression suite enumerates the full multivariate support, detects collisions, Fourier transforms exact small-circuit probabilities, checks active parity and $\mathbb{F}_2$ spans, verifies the two real-input counterexamples, compares the complete subset expansion against exact statevectors for one to four qubits, and validates global and lightcone-restricted bandwidths. The estimator tests independently compare sampled expectations and autodifferentiated gradients with exact circuits. These are internal correctness checks, not an external proof review.

C Primary-source novelty audit

The audit searched primary journal and arXiv records using combinations of "IQP", "anticommuting generator", "observable support", "lightcone", "affine input", "multidimensional Fourier", "product state", and "expectation estimator", then followed citations in the closest papers. It was not a systematic review of every thesis, patent, language, or citation descendant. That search limitation cannot rescue the claim: the located sources already establish enough of every candidate result to trigger the adverse criterion.

Gil Vidal and Theis give the $\{-1, 0, 1\}^n$ angle spectrum, multiaffine sine/cosine form, and signed sums from affine weights [1]. Schuld, Sweke, and Meyer give the general eigenvalue-difference Fourier theorem and its multidimensional form [2]. Casas and Cervera-Lierta analyze multidimensional degrees

of freedom [3]. Van den Nest supplies the general tractable-state/operator probabilistic estimation framework [4]. The recent IQP papers give the active anticommuting expansion, subset/XOR cancellation, Pauli and MMD estimators, and differentiation argument [5, 6]. The residual packaging and filter are useful, but do not satisfy the frozen central-novelty rule.

D Synthetic estimator resource-axis audit

The executable audit at `results/quantum-campaign-v1/evidence/estimator-scaling.json` is bound to the campaign manifest but excluded from every full-training-run pool. It uses deterministic synthetic angles, affine input maps, observable masks, and estimator seeds. The bounded-degree bank contains one-body generators and cycle edges; the Complete 2-local bank contains all one- and two-body generators. One-axis-at-a-time cases vary n, generator count, condition count K, and condition dimension d.

Every exact small-system record archives signed bias, maximum absolute bias, empirical variance, and RMSE for the expectation and phase-gradient estimator. At $n = 8$, where the audit does not claim an exact reference, it instead archives repeat-to-repeat RMSE and coordinatewise standard deviations. Each record also includes runtime, process peak RSS, input size and digest, repeat seeds, generator/active-generator counts, and the explicit interpretation that gradient magnitude is not a trainability proxy. Top-level provenance records the code revision and dirty-tree state, exact invocation, relevant source digests, Python/JAX/NumPy versions, operating system, backend, and device kind. The released 21-case archive has configuration digest `d647f220e3791c4eb678934d25f2ab1557e9396055f3b0e7a4a6ffdd04b28674` and deterministic numerical-results digest `54b5a75c383db7d605efe971d185734c00b2b47adcd0d5438cf6f2a77eeddfb0`. Wall time and memory are intentionally excluded from the latter digest because they depend on the execution host. The audit used zero full training runs.

E Frozen manifest and deterministic run plans

The production manifest has one canonical JSON digest and exact-key schema. It freezes the following scientific coordinates:

- coupling seeds $(666, 101, 202, 303, 404, 505, 606, 707)$ and aligned data seeds $(1666, 1101, 1202, 1303, 1404, 1505, 1606, 1707)$;
- optimizer seeds $(0, 1, 2, 3, 4)$;
- training betas $(0.15, 0.241\overline{6}, 1/3, 0.441\overline{6}, 0.55)$ and their four adjacent midpoints;
- 16 visible qubits and 136 Complete 2-local generators;
- bandwidths $(1, 2, 4)$, 2,000 Fourier operators per bandwidth, and 2,000 product-reference samples per operator and condition;
- eight MCMC chains, 4,000 declared burn-in sweeps, 32 declared thinning sweeps, 5,000 Training and 2,000 Validation samples per cell, and all sampler thresholds;
- balanced batches of 256, at most 2,000 Adam/cosine updates, Validation- only checkpoint selection, and nine bounded tuning candidates; and
- 10,000 paired hierarchical bootstrap replicates with seed 20260810 and all empirical gate thresholds.

The declared full-run identities comprise one runtime feasibility run, six separate-model representability screens, 36 tuning runs, one exact-objective diagnostic available only after representability failure,

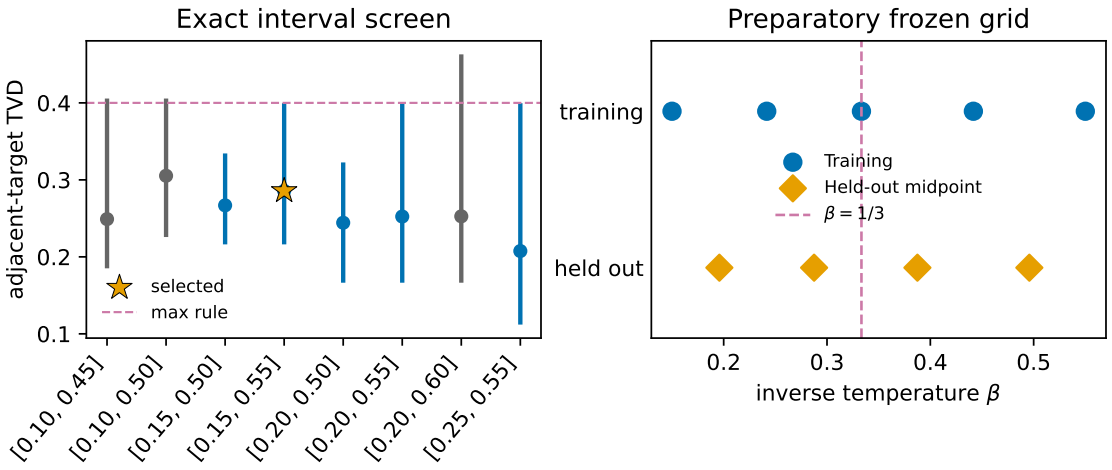


Figure 4: Preparatory exact target-only calibration. Left: minimum–maximum adjacent-target TVD and its median for each candidate interval; blue candidates passed every exact target rule, and the star marks the widest eligible interval. Right: five training betas and four adjacent held-out midpoints. The artifact is reproducible but was never released as a passed campaign gate.

320 primary runs, and 24 conditional ancilla runs. The feasibility/representability pool has a ceiling of 20; its undeclared remainder is intentionally unspendable. The phase tuning slice pairs the three preregistered feature maps one-to-one with the three declared learning-rate schedules, crossed with two pilot instances and two optimizer seeds. A primary or ancilla identity cannot be materialized until a passed tuning gate contains an immutable evidence-bound decision.

The primary 320 count is

$$8 \text{ instances} \times 5 \text{ seeds} \times (3 \text{ shared} + 5 \text{ separate}) = 320. \tag{40}$$

The ancilla 24 count is

$$2 \text{ instances} \times 3 \text{ seeds} \times 2 \text{ rungs} \times (1 \text{ ancilla} + 1 \text{ control}) = 24. \tag{41}$$

Unused identities in one budget pool cannot be reserved in another.

F Ising reconstruction and preparatory calibration

The public PennyLane generator uses a periodic NetworkX square grid, no local fields, legacy NumPy seed 666, directed independent edge draws in $[0, 2]$, and arithmetic symmetrization. The reconstructed seed-666 coupling matrix has SHA-256

`58cd62927e3be95a827e98efb7282af5a62d2232e15953da6362599017989194`.

Scoring the released 55,000 strings at $T = 3$ under this matrix gives mean energy -18.7718575999; exact enumeration gives -18.7697295208, a difference of -2.20649×10^{-4} exact energy standard deviations. This joint check protects the coupling order, sign, factor of one half, bit-to-spin map, and $\beta = 1/T$ convention.

Figure 4 summarizes the model-free screen. The selected interval has minimum adjacent TVD 0.2162, maximum adjacent TVD 0.3996, minimum normalized entropy 0.1874, and maximum point probability 0.3964, meeting the frozen limits 0.015, 0.40, 0.18, and 0.40. Its calibration JSON has SHA-256

`8d121e4a519766ff0234e00021815b3bcfd740e9fc9284dbf5b2aa448f29f6ae`.

This matching preparatory artifact cannot pass its gate after a failed predecessor.

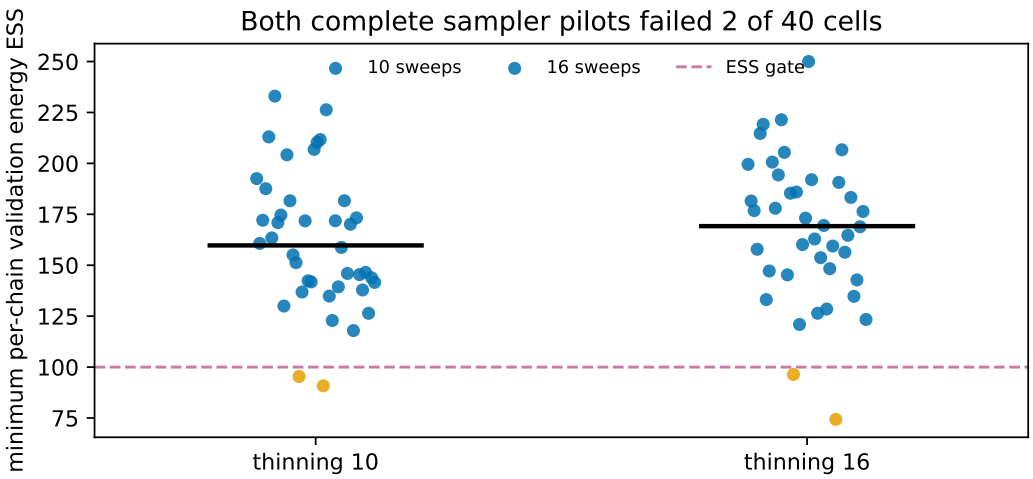


Figure 5: Complete sampler-development attempts. Each point is the minimum energy ESS over the eight Validation chains in one instance/beta cell; orange points miss the threshold. Both schedules fail 2 of 40 cells, so neither is a certified campaign dataset.

G Sampler-development audit

The frozen distributional sampler criterion is exact-reference RMSE over the declared complete bank of one- and two-body parity characters, with threshold 0.08. The audit also archives plug-in empirical Bitstring TVD over all 65,536 states, but does not gate on it: at 7,000 retained Training-plus-Validation strings its unseen-cell mass is dominated by finite-support undersampling. The character rule is a disclosed low-order/Fourier distributional diagnostic, not a claim of certified full-law TVD.

The zero-field target is invariant under a global spin flip. The PennyLane-style single-spin Metropolis kernel was therefore augmented by an independent global flip of each retained state. This always-accepted sampler-only transition preserves the exact law and avoids falsely treating a trapped ferromagnetic sector as model error. Every attempt records its chain count, burn-in, thinning, proposal count, symmetry transition, seeds, hashes, and exact-reference diagnostics [13].

The audit also checks maximum chain-energy deviation, pooled energy error, magnetization, pair-correlation RMSE, and weight-one/two character RMSE against the exactly enumerated target. In the two complete attempts, only the ESS rule failed. A declared 4,000-burn-in/32-thinning attempt was then started, but the novelty verdict arrived after six cells and the process was interrupted with exit 130. The partial files and interruption note are retained. No complete summary exists, no sampling-quality gate passed, and no model consumed the data.

H Binding statistical rules

Had the primary matrix been released, coupling instances—not optimizer seeds or conditions—would have been the primary scientific replicates. Each of 10,000 bootstrap draws would first resample eight instances with replacement, then resample the five matched optimizer seeds within each selected instance, preserving model and condition pairings. The frozen seed is 20260810 and the interval is the 95% percentile interval.

For a held-out cell, define the benefit retention

$$r = \frac{\mathrm{TVD}_{\mathrm{no}} - \mathrm{TVD}_{\mathrm{fixed}}}{\mathrm{TVD}_{\mathrm{no}} - \mathrm{TVD}_{\mathrm{phase}}}. \tag{42}$$

The gate requires median $r \geq 0.5$ and a strictly positive lower bootstrap bound. Missing methods, nonpositive denominators, or incompatible pairings fail conservatively.

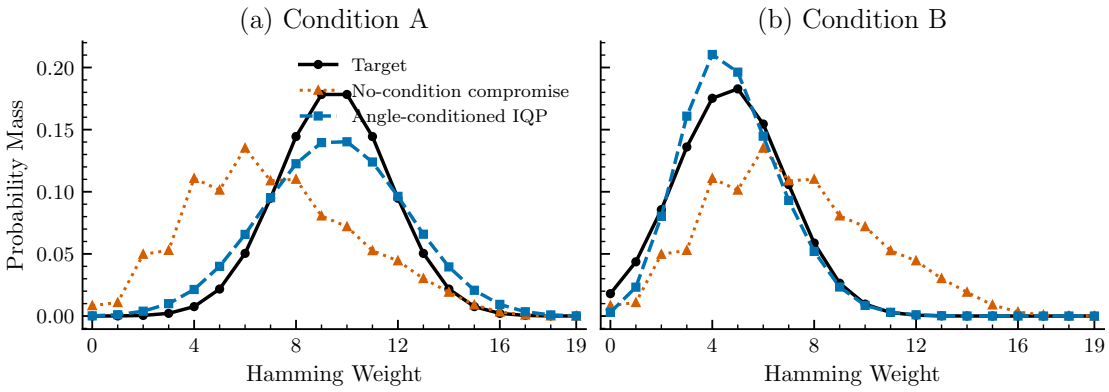


Figure 6: Legacy Gaussian motivation, regenerated from its archived ten-seed study. One shared angle-conditioned checkpoint family changes its Hamming- weight profile with the input; the no-condition output compromises. This is not Ising evidence and does not override the failed novelty gate.

For each held-out midpoint, the Oracle adjacent-beta lookup evaluates the two neighboring stored training outputs against the exact held-out target and keeps the lower TVD. Fixed input must have a strictly positive paired improvement interval, win at least 75% of instance-by-held-out-condition cells, and worsen no instance's mean. All three components are binding. The ancilla rule similarly requires a preregistered rung to beat both $m = 0$ and its matched-capacity visible control in paired median, 75% cell wins, and both instance guardrails; both $m = 3$ and $m = 5$ would be reported. No values for any of these rules exist because their predecessor stages were blocked.

I Legacy Gaussian motivation and excluded studies

Before the Ising campaign was designed, an archived 19-qubit Complete 2-local study trained one angle-conditioned model on two permutation-invariant Gaussian Hamming-weight targets. Its ten optimizer seeds are the replication units. Figure 6 illustrates the profile response; the exact mean-condition profile and Bitstring TVDs were 0.096 and 0.314 for angle conditioning, 0.365 and 0.482 for no condition, and 0.364 and 0.592 for a basis-condition control. Separate fits were capacity references. The remaining Bitstring error exposed nonuniformity within Hamming shells and motivated the untwirled, structured Ising benchmark.

Historical sparse-grid, condition-count, topology, ancilla, and negative audits remain archived in the repository. A target-specific permutation twirl can make Bitstring TVD equal profile TVD for a known distribution that is uniform within Hamming shells, but it changes the output protocol and does not show that the untwirled circuit learned the symmetry. Those results are not part of the present central evidence line and cannot rescue the blocked Ising campaign.

J Reproducibility inventory

The package provides the following independently inspectable layers:

1. a strict JSON manifest whose digest binds model, data, estimator, training, statistical, and budget rules;
2. exact Ising coupling reconstruction, target enumeration, calibration, and content-addressed target archives;
3. immutable gate records, reservations, result-bundle schemas, and monotonic stage release;
4. deployment signatures, phase-feature maps, role-tagged datasets, balanced training helpers, and exact-TV evaluation;

5. hierarchical bootstrap, benefit-retention, Oracle, interpolation, and ancilla decision functions;
6. theorem-support, estimator-scaling, response-spectrum, campaign, and manuscript audit tests; and
7. deterministic scripts for evidence figures, report tables, the main manuscript, and this supplement.

The central regeneration commands are:

```
uv sync --group dev
uv run pytest
uv run ruff check src scripts tests
uv run python scripts/prepare_ising_campaign.py --targets-only
uv run python scripts/run_ising_campaign.py --stage primary --dry-run
uv run python scripts/generate_quantum_draft_figures.py
uv run python scripts/run_estimator_scaling.py \
  --output /tmp/iqp-cond-estimator-scaling.json
uv run python scripts/audit_response_spectrum.py \
  --output /tmp/iqp-cond-response-spectrum.json  # blocked, exit 2
uv run python scripts/build_campaign_report.py
(cd paper && latexmk -pdf -interaction=nonstopmode \
  -halt-on-error main.tex)
```

The target-only command rechecks existing exact archives and releases no gate. The first theorem gate is already terminally failed. Running either the full preparation command or a downstream campaign CLI command returns a blocked status before dataset generation, reservation, or trainer access. The repository does not contain or claim persistent Ising checkpoints.

K Adversarial review checklist

Before any future external circulation, an independent quantum-information or QML expert must review the proofs, closest-prior search, estimator scope, and significance. A new positive decision must not edit the existing failed gate; it requires a new manifest version and evidence chain. The following items also remain mandatory:

- replace anonymous metadata and generic contribution/interest statements;
- validate every DOI and arXiv version against the final bibliography;
- rerun the complete suite and clean-environment regeneration;
- inspect every rendered page for blanks, clipping, unresolved references, and overfull boxes; and
- archive independent review responses and allow major objections to reopen the theorem, claims, experiments, or venue decision.

Data and code availability

The repository contains the frozen manifest, campaign/ledger implementation, exact Ising target construction, target-only calibration, failed sampler attempts, theorem dossier and checks, statistical rules, analysis scripts, and the complete manuscript sources. Large public-source Ising data are used only for a reconstruction check; final target truth is generated by exact enumeration. No trained Ising checkpoints exist because the novelty gate failed.

Author contributions and generative-AI use

The anonymous author(s) designed the study, derived and checked the analysis, implemented the software, audited the evidence, and wrote the manuscript. This generic internal statement must be replaced with named CRediT-style contributions before any submission. OpenAI Codex assisted with software implementation, literature organization, test generation, figure and manuscript drafting, and consistency checks. The human author(s) remain responsible for verifying every proof, citation, artifact, and statement before external circulation. No AI system is listed as an author.

Competing interests

No competing-interest declaration has been supplied for this anonymous internal draft. A verified author declaration is required before submission.

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